

# Final Report

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## Abstract

## Introduction and Literature Review

This report is structured as follows

## Methods

Data from 1960 to 2024 on Total Live Births (TLB) and Total Fertility Rates (TFR) was pulled from Singapore Department of Statistics (data.gov.sg). The data was then transformed into a tsibble object with three columns: Year, Total Fertility Rates (TFR) and Total Live Births.

The TFR was initially plotted to examine the presence of trends, seasonality, and other notable patterns (Figure 1). The subsequent analysis focuses on the period from 1960 to 2012, a second plot was produced for visualisation and modelling purposes (Figure 2).

There is a long-term decreasing trend over time hence the data was differenced. Viable ARIMA models were found but the KPSS unit root test suggested the data needed to be differenced again where more viable ARIMA models were found. The initial plot decreasing trend suggests a log transformation would be appropriate where more models were found after differencing and twice differencing.

Figure 1. Initial Plot of Total Fertility Rate (TFR)

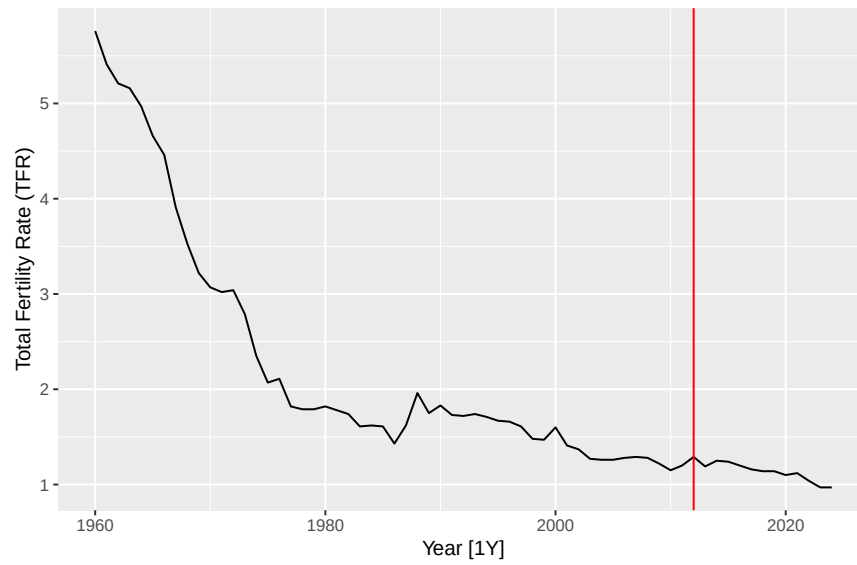
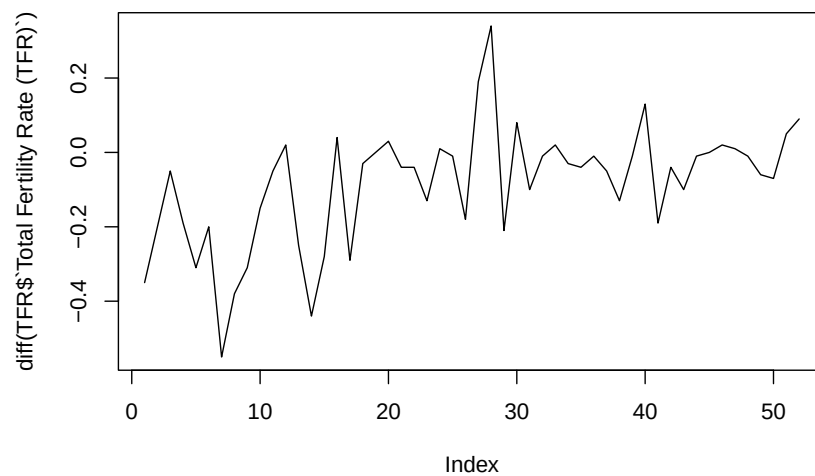
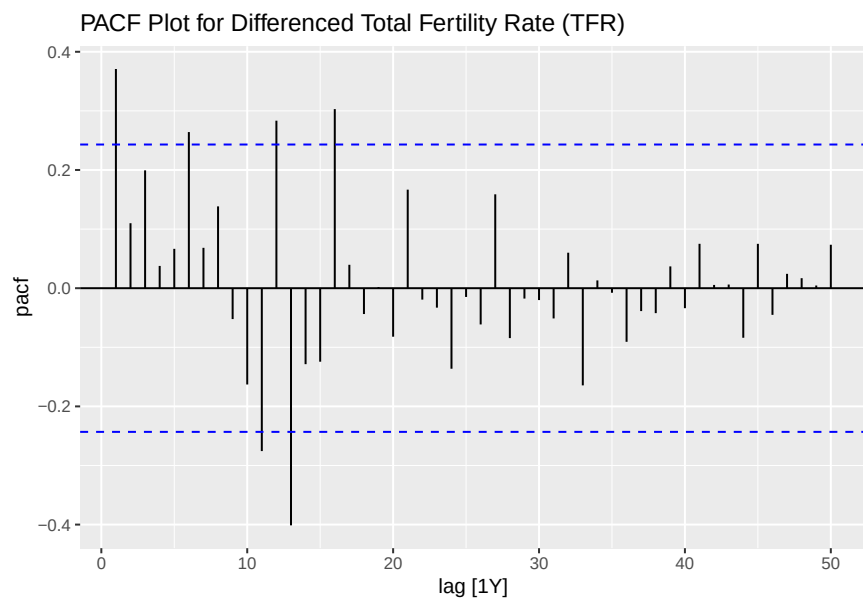
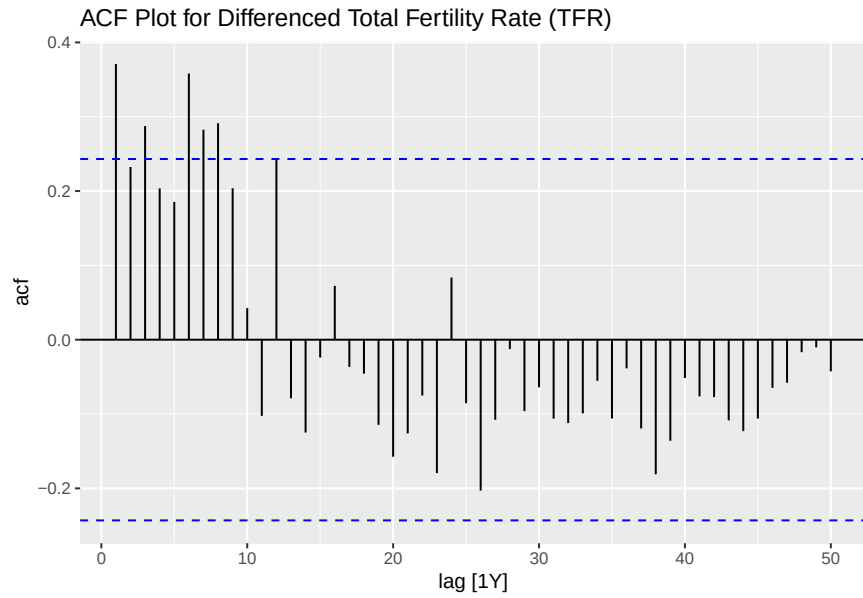


Figure 1 show a long-term decreasing trend over time

**Plot of Differenced Total Fertility Rate (TFR) until 2012**

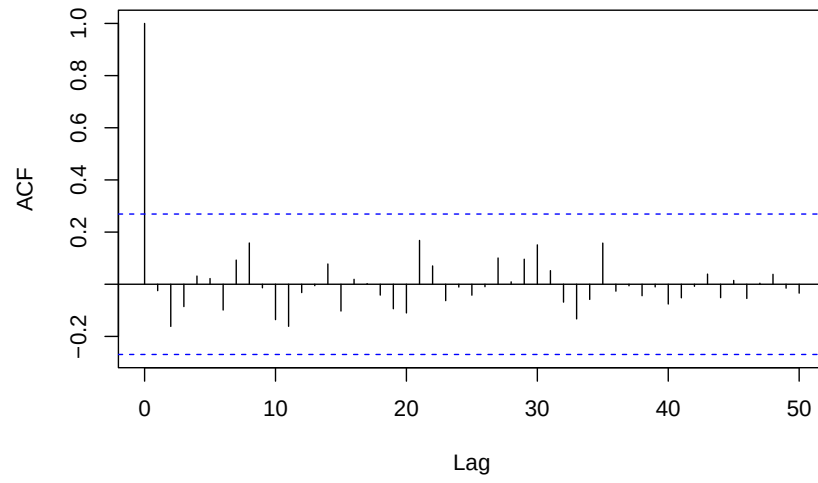


```
##      kpss_stat kpss_pvalue
##      0.8412374  0.0100000
```

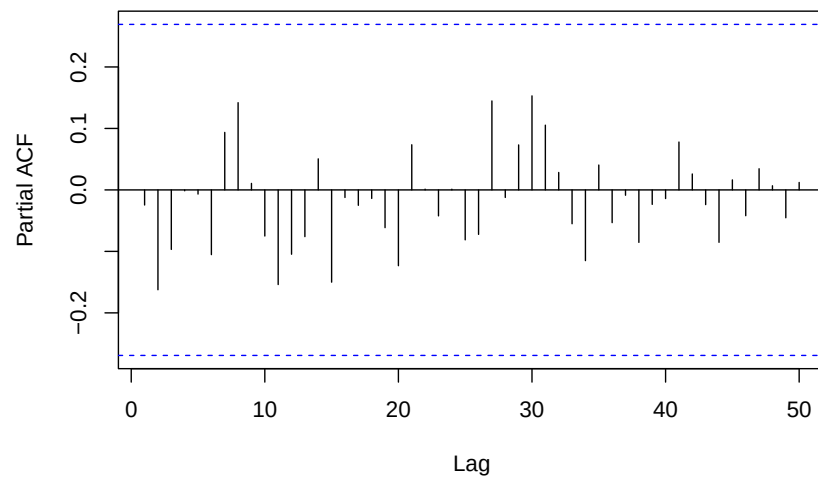


```
## Series: TFR_ts
## ARIMA(16,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.3647 -0.1061  0.4577 -0.2586  0.2984  0.2452  0.2006  0.1578 -0.0293
## s.e.  0.1240  0.1289  0.1246  0.1128  0.0935  0.0832  0.0952  0.0928  0.0996
##      ar10     ar11     ar12     ar13     ar14     ar15     ar16
##      -0.0306 -0.3594  0.3889 -0.6048 -0.0419 -0.3361  0.5135
## s.e.  0.0973  0.0913  0.1039  0.1066  0.1350  0.1295  0.1397
##
## sigma^2 = 0.009716: log likelihood = 48.99
## AIC=-63.99  AICc=-45.99  BIC=-30.82
```

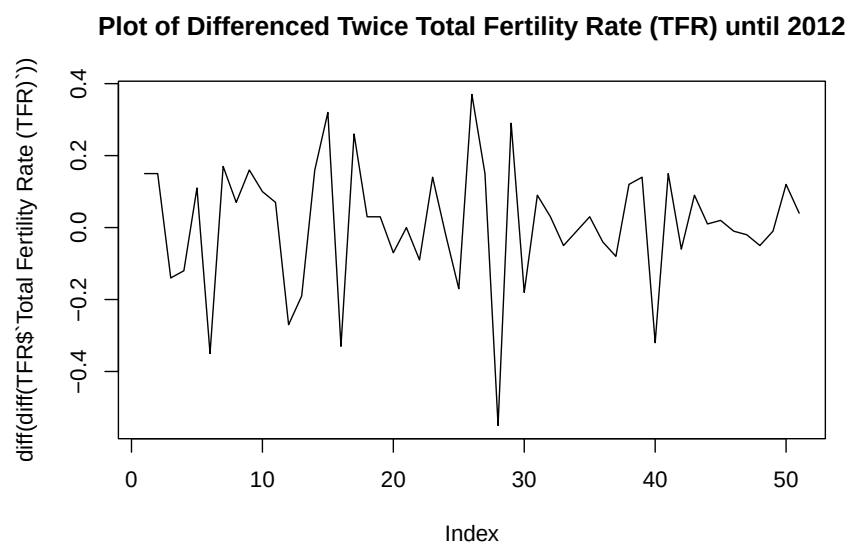
**ACF of Residuals for ARIMA(16,1,0) for TFR**



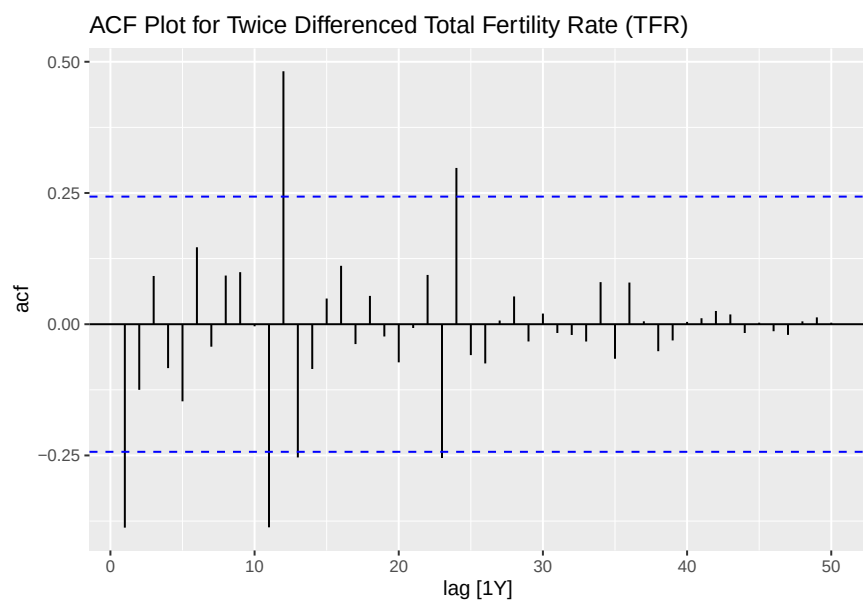
**PACF of Residuals for ARIMA(16,1,0) for TFR**

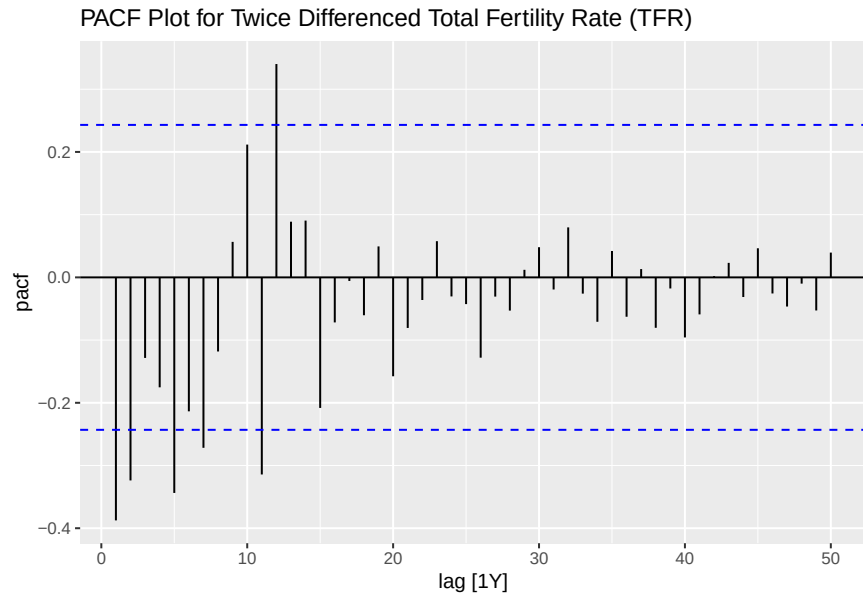


By using differencing, the plots look like the trend has been removed.



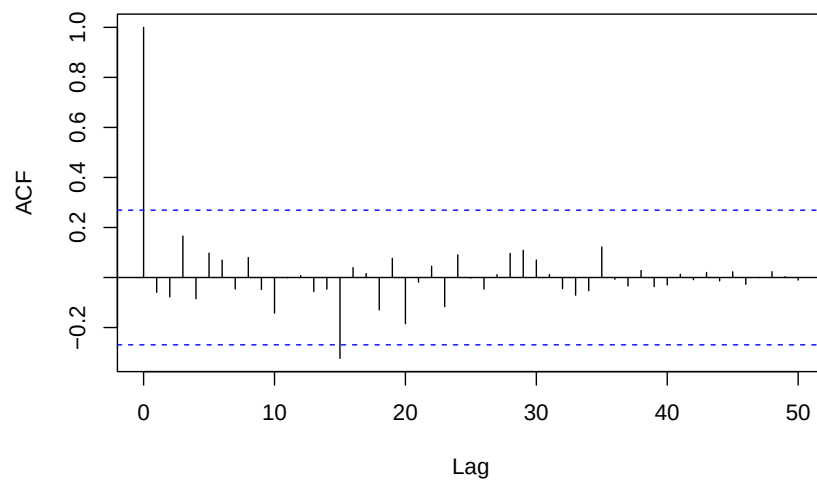
```
##      kpss_stat kpss_pvalue
## 0.04262877 0.10000000
```



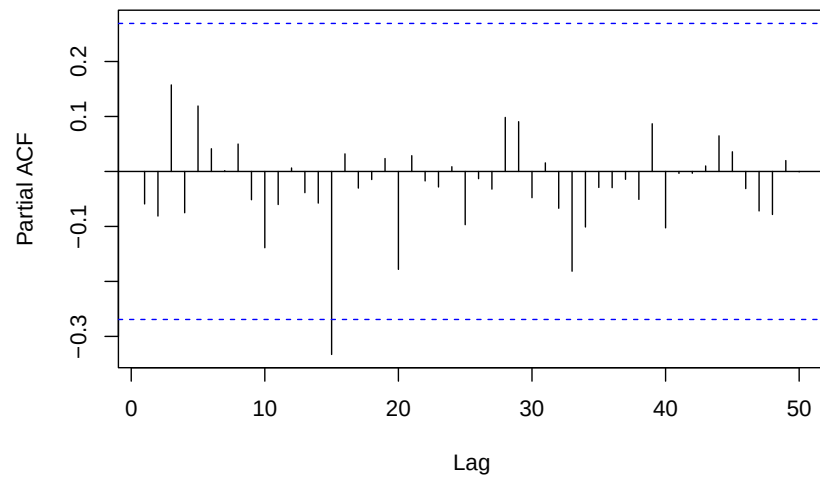


```
## Series: TFR_ts
## ARIMA(12,2,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      -0.6234 -0.7441 -0.5813 -0.5700 -0.4163 -0.1741  0.0462  0.3482
## s.e.   0.1203  0.1494  0.1673  0.1676  0.1705  0.1772  0.1846  0.1847
##      ar9      ar10     ar11     ar12
##      0.4206  0.4880  0.1609  0.5040
## s.e.   0.1845  0.1783  0.1540  0.1205
##
## sigma^2 = 0.01286: log likelihood = 41.78
## AIC=-57.55  AICc=-47.72  BIC=-32.44
```

ACF of Residuals for ARIMA(12,2,0) for TFR



**PACF of Residuals for ARIMA(12,2,0) for TFR**

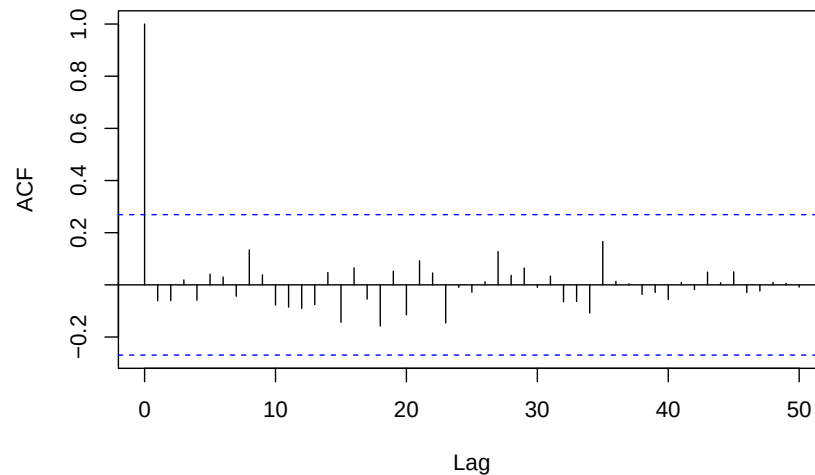


```

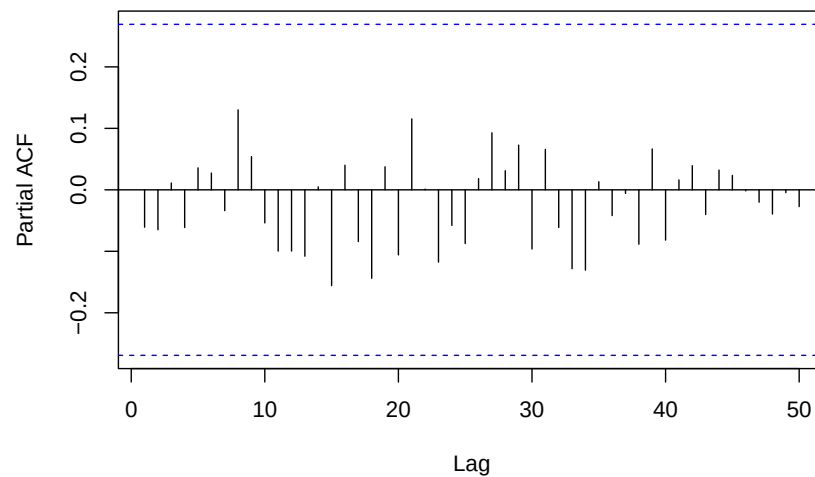
## Series: TFR_ts
## ARIMA(12,2,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.4622 -0.7606 -0.9273 -0.8808 -0.6967 -0.4124 -0.2312  0.0490
## s.e.   0.1091  0.1266  0.1643  0.2032  0.2376  0.2529  0.2548  0.2404
##      ar9      ar10     ar11     ar12     ma1      ma2      ma3
##     0.1578  0.3268  0.1426  0.6463 -0.1402  0.1411  0.8593
## s.e.   0.2067  0.1645  0.1248  0.1005  0.1237  0.1262  0.1309
##
## sigma^2 = 0.009277: log likelihood = 47.29
## AIC=-62.57  AICc=-46.57  BIC=-31.66

```

**ACF of Residuals for ARIMA(12,2,3) for TFR**

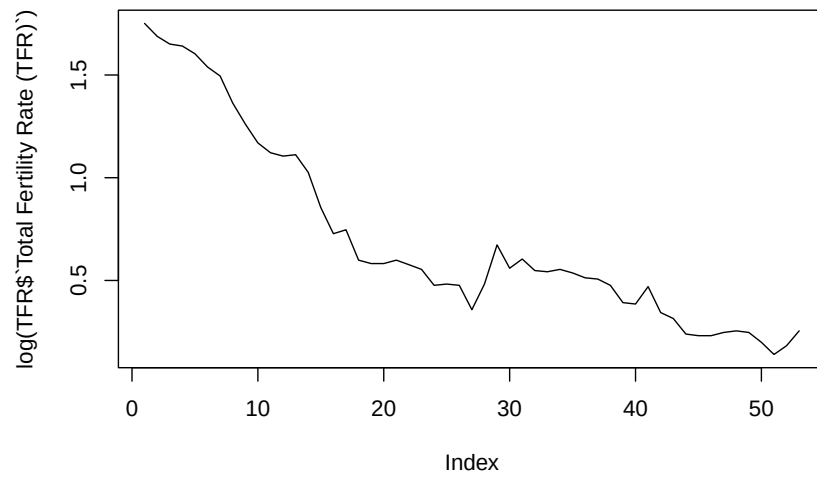


**PACF of Residuals for ARIMA(12,2,3) for TFR**

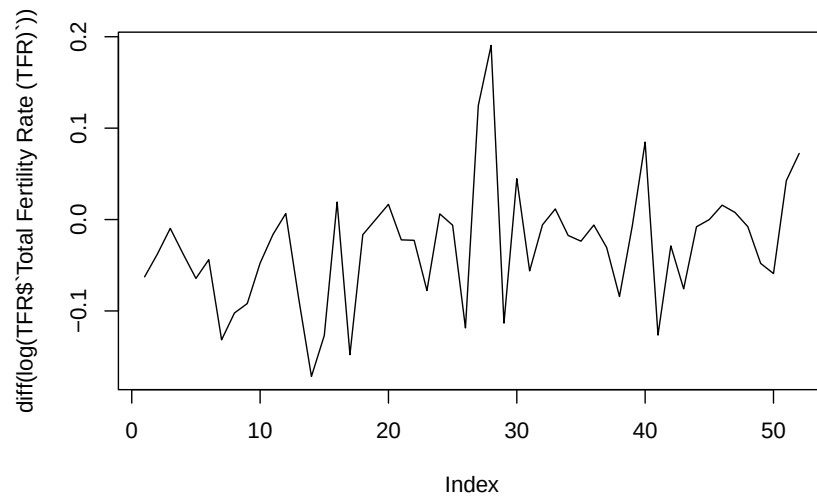




**Plot of Log Transformation for Total Fertility Rate (TFR) until 2012**

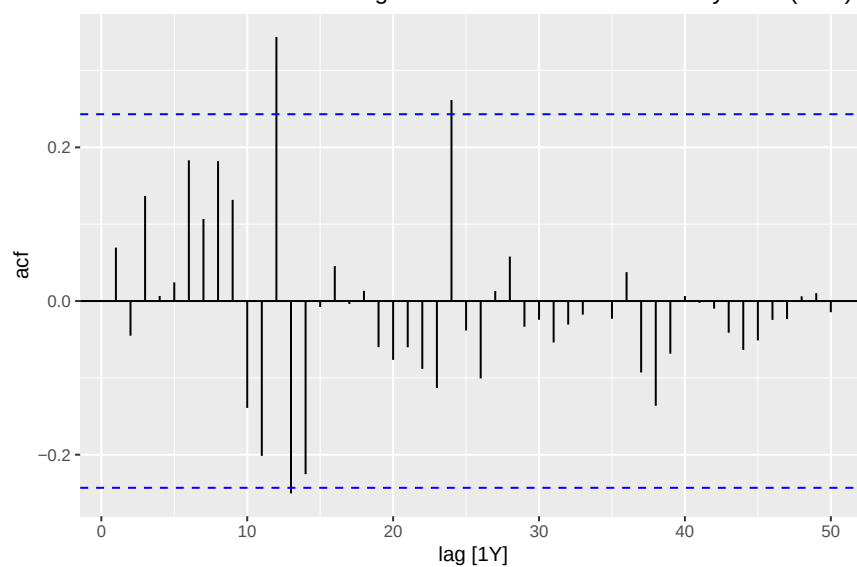


**Plot of Differenced Log Transformation for Total Fertility Rate (TFR) until 2012**

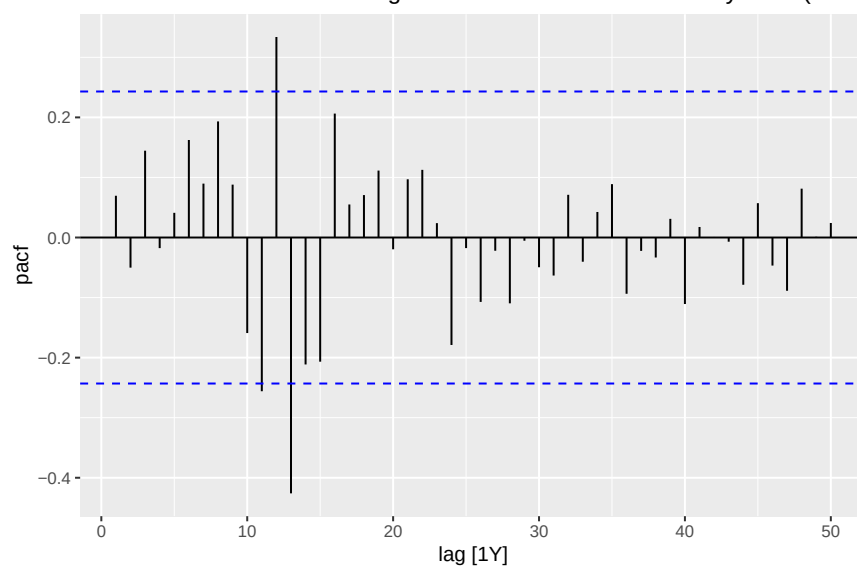


```
##      kpss_stat kpss_pvalue
## 0.50484854 0.04057465
```

ACF Plot for Differenced Log Transformation of Total Fertility Rate (TFR)

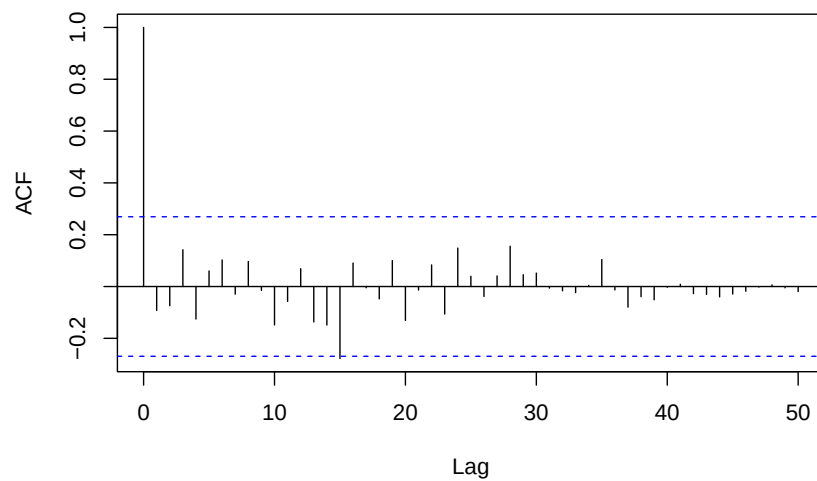


PACF Plot for Differenced Log Transformation of Total Fertility Rate (TFR)

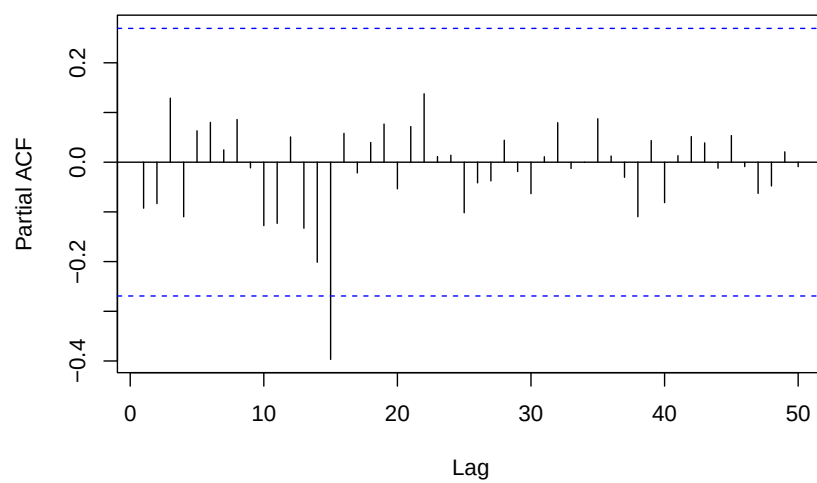


```
## Series: log(TFR_ts)
## ARIMA(13,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.2629 -0.0799  0.1194  0.0024  0.1349  0.1715  0.1679  0.2392  0.0298
## s.e.  0.1215  0.1145  0.1065  0.1064  0.1079  0.0974  0.1023  0.1036  0.1106
##      ar10     ar11     ar12     ar13
##      -0.0307 -0.2349  0.3587 -0.4375
## s.e.  0.1130  0.1084  0.1170  0.1258
##
## sigma^2 = 0.00323: log likelihood = 79.9
## AIC=-131.8  AICc=-120.45  BIC=-104.48
```

**ACF of Residuals for ARIMA(13,1,0) for TFR with log transformation**

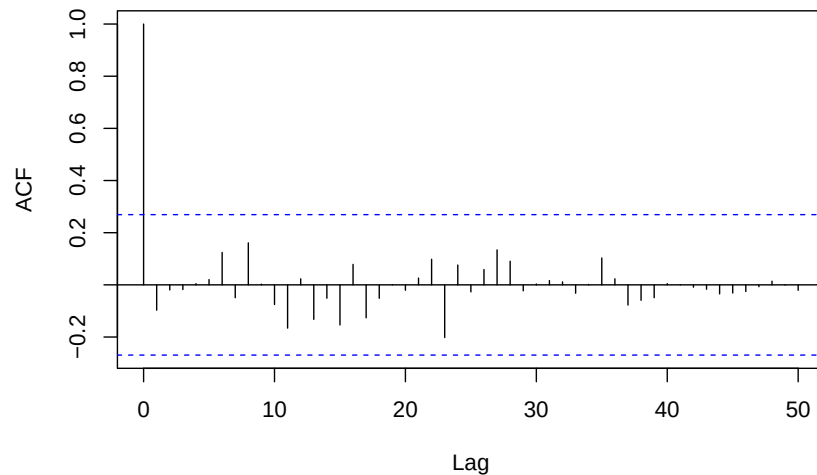


**PACF of Residuals for ARIMA(13,1,0) for TFR with log transformation**

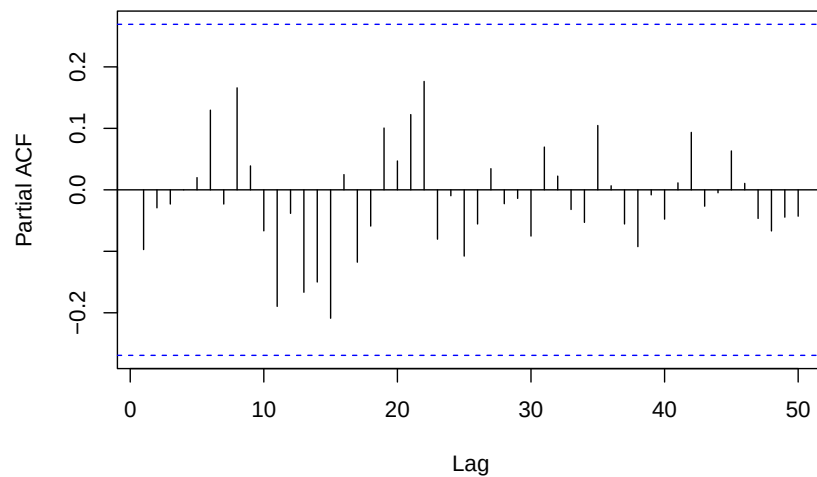


```
## Series: log(TFR_ts)
## ARIMA(13,1,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.2783 -0.2434 -0.2961 -0.0013  0.1400  0.2218  0.1567  0.2497  0.0839
## s.e.  0.1256  0.1019  0.1096  0.1094  0.1091  0.0975  0.1024  0.1087  0.1147
##      ar10     ar11     ar12     ar13     ma1      ma2      ma3
##      0.0722 -0.0719  0.4674 -0.5235  0.0903  0.1376  0.9749
## s.e.  0.1170  0.1097  0.1005  0.1256  0.0271  0.1085  0.1089
##
## sigma^2 = 0.00232:  log likelihood = 85
## AIC=-136   AICc=-118   BIC=-102.83
```

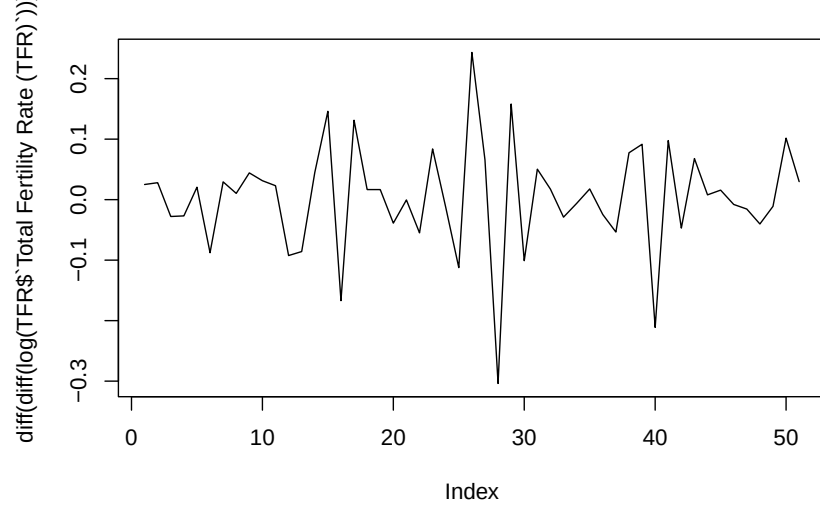
**ACF of Residuals for ARIMA(13,1,3) for TFR with log transformation**



**PACF of Residuals for ARIMA(13,1,3) for TFR with log transformation**

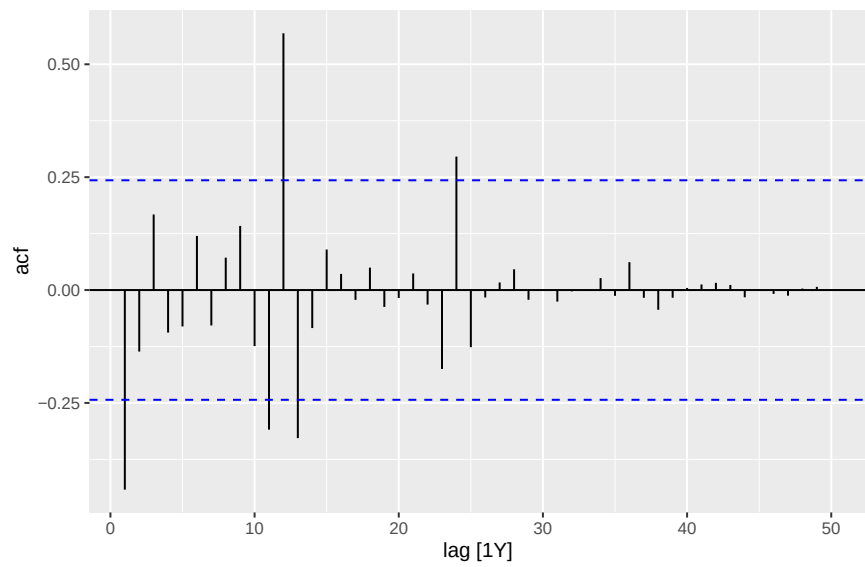


**Plot of Differenced Twice Log Transformation for Total Fertility Rate (T**

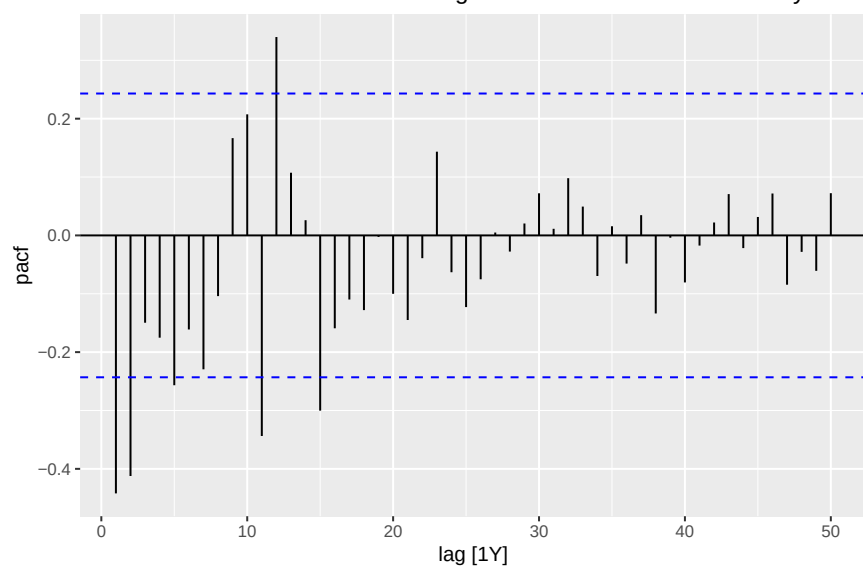


```
##      kpss_stat kpss_pvalue
## 0.04760604 0.10000000
```

**ACF Plot for Twice Differenced Log Transformation of Total Fertility Rate (**

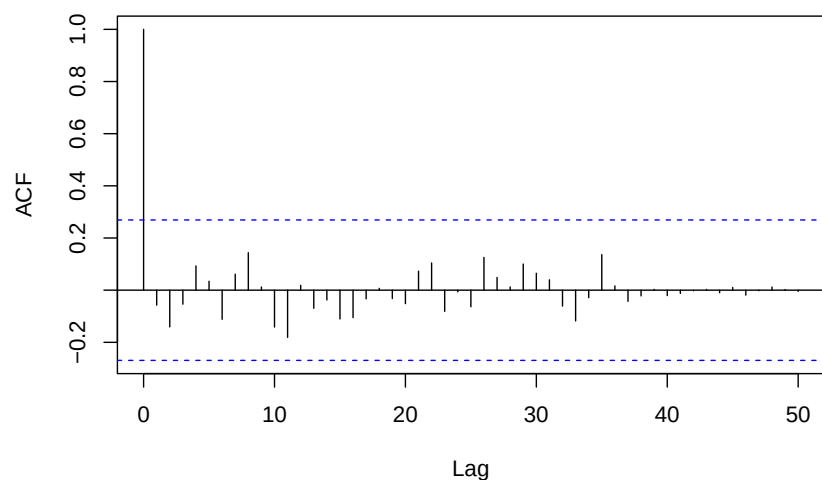


PACF Plot for Twice Differenced Log Transformation of Total Fertility Rate (

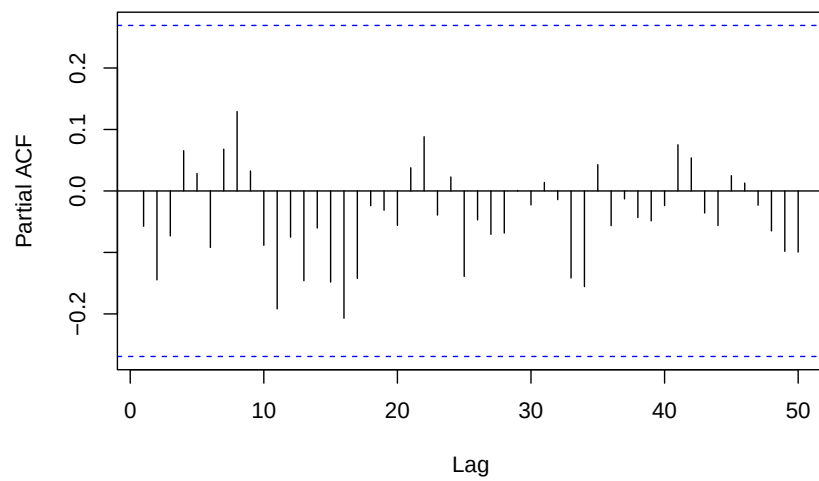


```
## Series: log(TFR_ts)
## ARIMA(15,2,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      -0.7137 -0.7004 -0.3372 -0.5826 -0.3331 -0.1103  0.0628  0.2087
## s.e.   0.1218  0.1571  0.1913  0.1824  0.2031  0.2111  0.2146  0.2149
##      ar9      ar10     ar11     ar12     ar13     ar14     ar15
##      0.1864  0.1142 -0.1162  0.3260 -0.1661 -0.2596 -0.4876
## s.e.   0.2152  0.2137  0.2042  0.1855  0.1901  0.1618  0.1289
##
## sigma^2 = 0.002602: log likelihood = 82.83
## AIC=-133.67  AICc=-117.67  BIC=-102.76
```

ACF of Residuals for ARIMA(15,2,0) for TFR with log transformation



**PACF of Residuals for ARIMA(15,2,0) for TFR with log transformatio**

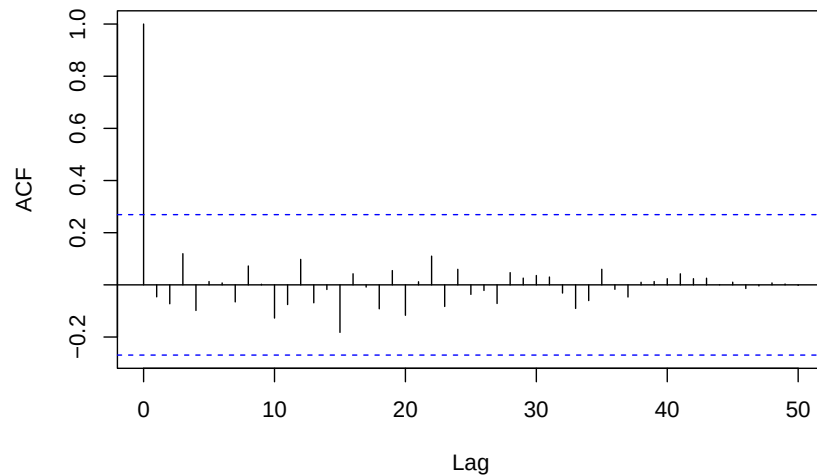


```

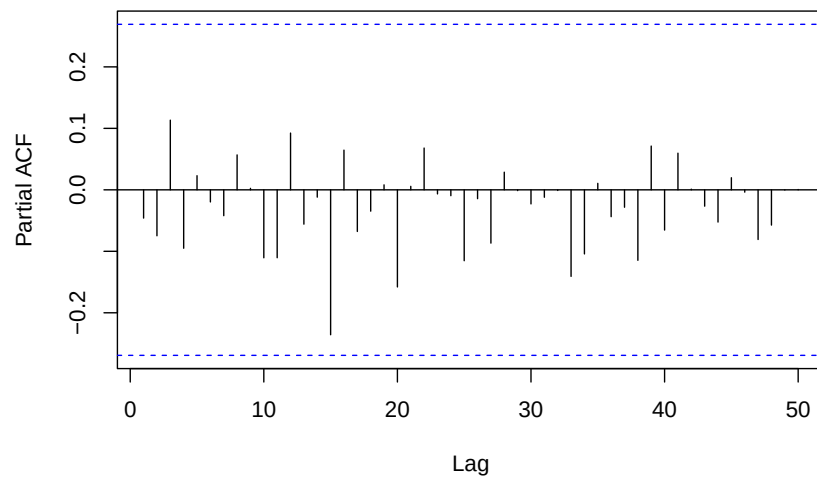
## Series: log(TFR_ts)
## ARIMA(13,2,2)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      0.6514 -0.2412 -0.0063 -0.1608  0.0605  0.0715  0.0280  0.1159
## s.e.  0.1194  0.1202  0.1193  0.1180  0.1158  0.1095  0.1099  0.1125
##      ar9      ar10     ar11     ar12     ar13      ma1      ma2
##      -0.1105 -0.0649 -0.1966  0.4240 -0.6274 -1.6346  0.7851
## s.e.   0.1156   0.1159   0.1124  0.1126  0.0987   0.1446  0.1392
##
## sigma^2 = 0.002717: log likelihood = 82.19
## AIC=-132.39  AICc=-116.39  BIC=-101.48

```

**ACF of Residuals for ARIMA(13,2,2) for TFR with log transformation**



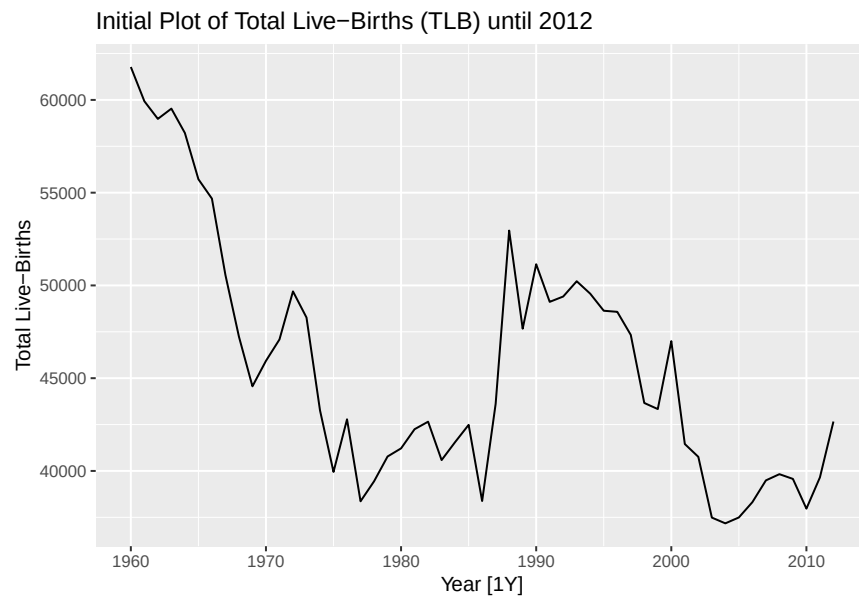
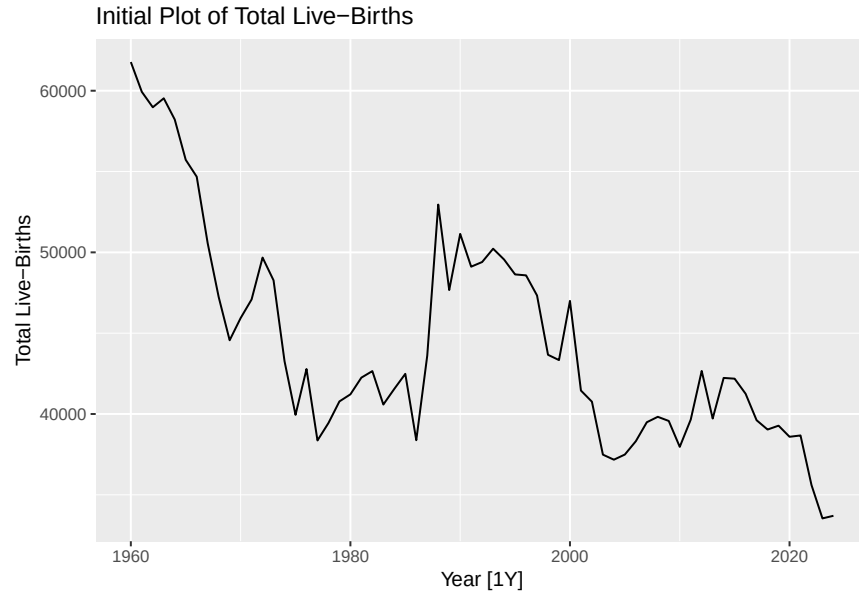
**PACF of Residuals for ARIMA(13,2,2) for TFR with log transformation**





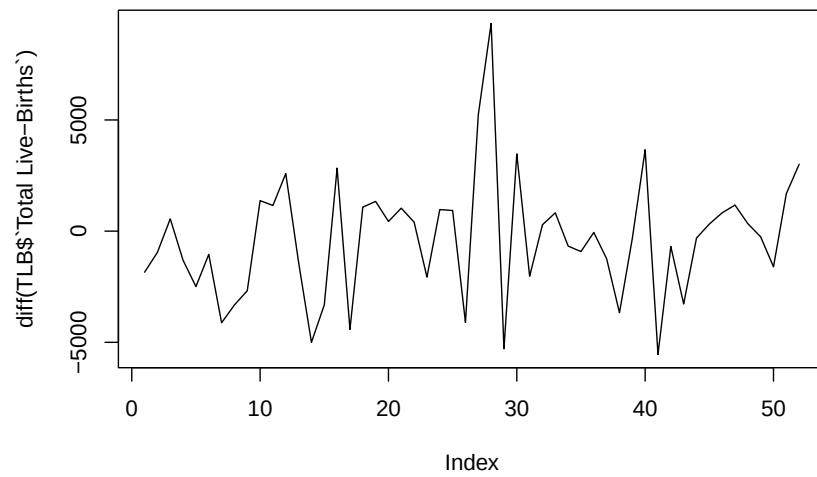
The ACF for TFR shows significant positive autocorrelations from lags 1 to 9, while TLB shows significant positive autocorrelations from lags 1 to 6, with a slow decrease in the ACF as the lag increases. This suggests that neither series is white noise, as there are multiple spikes outside the significance bounds.

The presence of a strong trend and slow decrease confirms that both TFR and TLB are non-stationary series. The absence of a sharp cut-off suggests that differencing or an autoregressive (AR) model may be suitable for this series. An extended ACF plot for TFR confirms that significant positive autocorrelations persist up to lag 10.



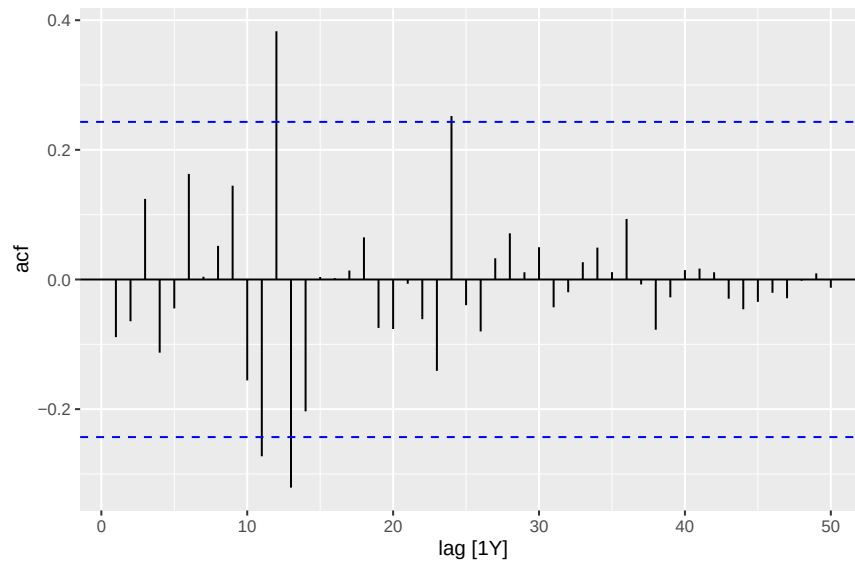
From the initial plots of TFR and TLB, both series show a long-term decreasing trend over time, with TFR exhibiting a more pronounced decline. A notable change is observed around 1980 to 1990, where both series display a temporary broad peak before continuing their downward trend. As the data is yearly, seasonality cannot be determined. While both series may suggest cyclical behaviour, the trend would need to be removed to confirm this.

**Plot of Differenced Total Live Births (TLB) until 2012**

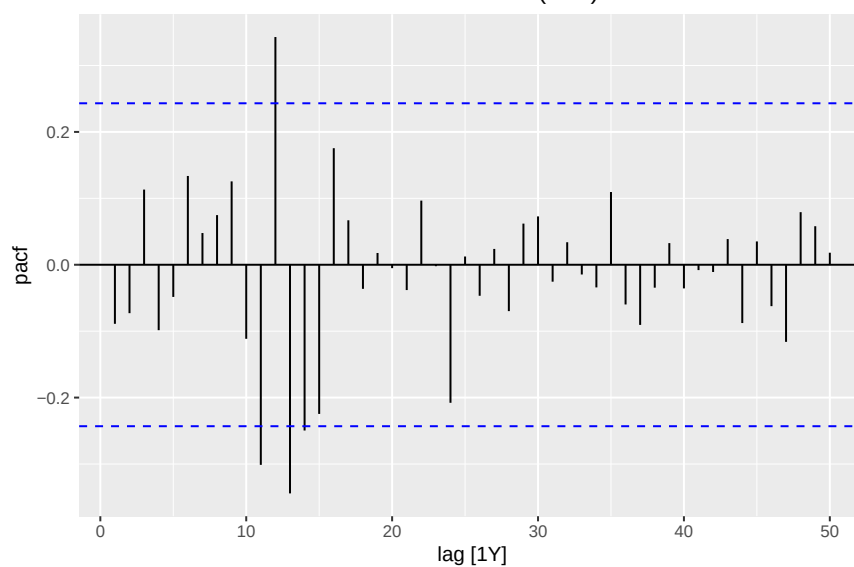


```
##      kpss_stat kpss_pvalue  
## 0.2126452    0.1000000
```

**ACF Plot for Differenced Total Live Births (TLB)**

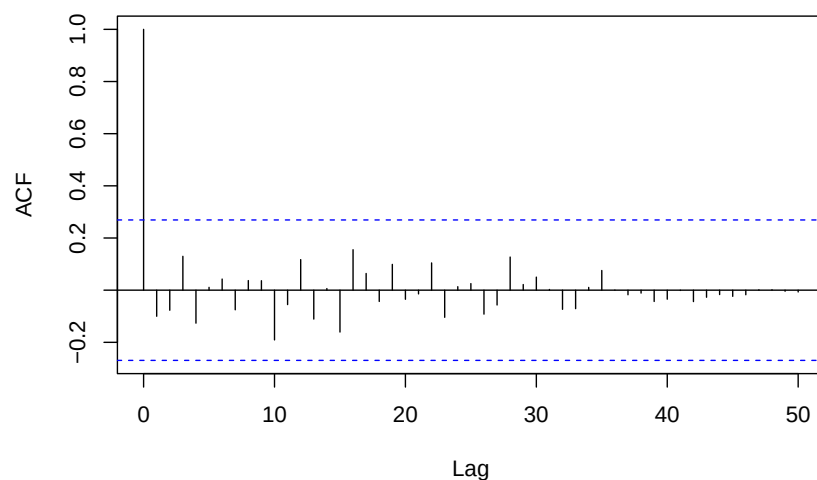


PACF Plot for Differenced Total Live Births (TLB)

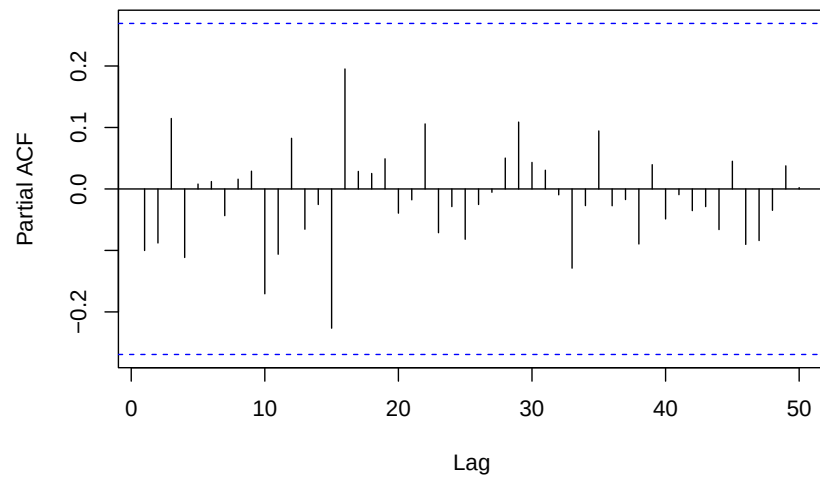


```
## Series: TLB_ts
## ARIMA(14,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.0260  0.0380 -0.0184 -0.0555  0.0794  0.1459  0.1111  0.1666  0.0608
## s.e.  0.1344  0.1218  0.1091  0.1027  0.1050  0.1015  0.1087  0.1051  0.1087
##      ar10     ar11     ar12     ar13     ar14
##      -0.0696 -0.2211  0.2823 -0.3715 -0.2759
## s.e.   0.1085   0.1078  0.1195  0.1245  0.1466
##
## sigma^2 = 5629775:  log likelihood = -472.66
## AIC=975.33   AICc=988.66   BIC=1004.6
```

ACF of Residuals for ARIMA(14,1,0) for TLB



**PACF of Residuals for ARIMA(14,1,0) for TLB**

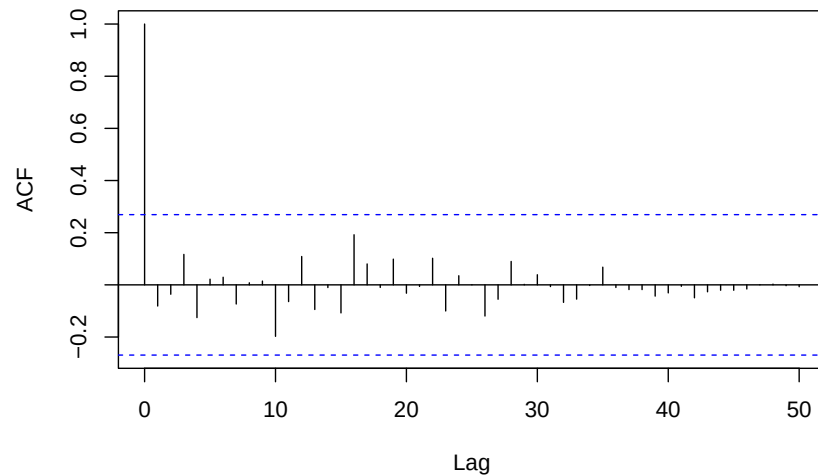


```

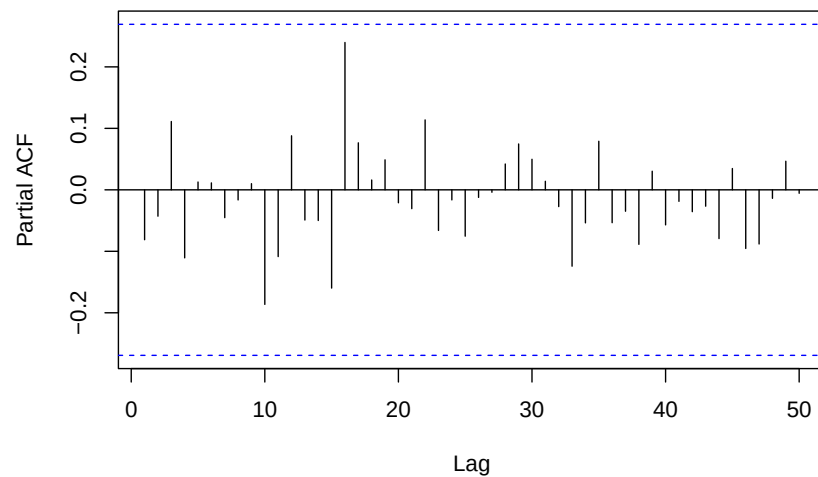
## Series: TLB_ts
## ARIMA(13,1,1)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.5571 -0.0703  0.0264 -0.0762  0.0940  0.1004  0.0542  0.1077 -0.0216
## s.e.  0.1677  0.1156  0.1116  0.1120  0.1104  0.1040  0.1090  0.1095  0.1125
##      ar10     ar11     ar12     ar13     ma1
##      -0.0853 -0.1964  0.4091 -0.5605 -0.5344
## s.e.  0.1130  0.1112  0.1223  0.1126  0.1886
##
## sigma^2 = 5503186: log likelihood = -472.28
## AIC=974.57  AICc=987.9  BIC=1003.83

```

**ACF of Residuals for ARIMA(13,1,1) for TLB**



**PACF of Residuals for ARIMA(13,1,1) for TLB**

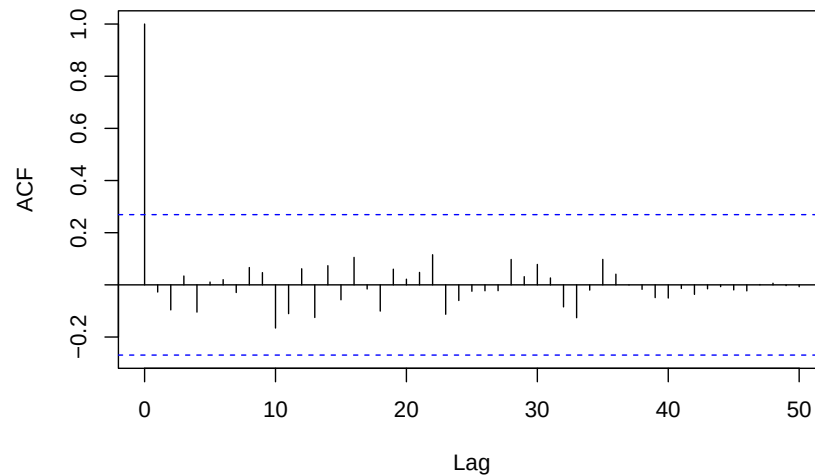


```

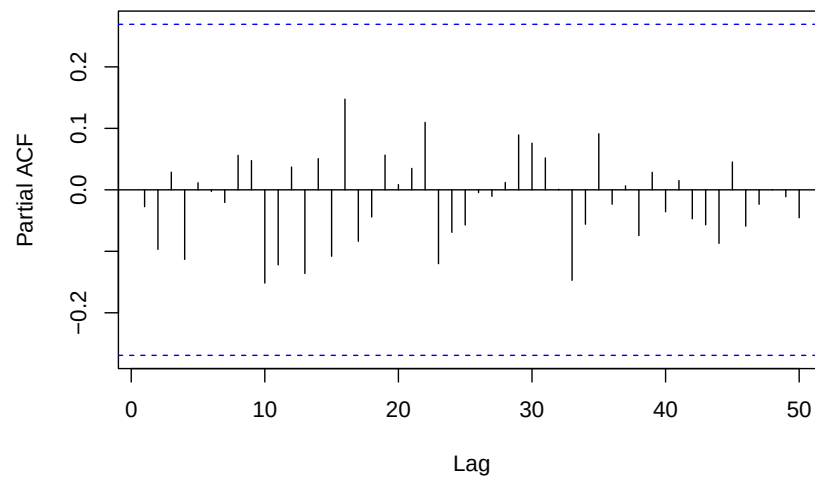
## Series: TLB_ts
## ARIMA(13,1,2)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      0.6648 -0.4027 -0.0077 -0.0835  0.1408  0.0861  0.0430  0.1468
## s.e.  0.1376  0.1968  0.1260  0.1252  0.1252  0.1203  0.1199  0.1246
##      ar9      ar10     ar11     ar12     ar13      ma1      ma2
##      -0.0053 -0.0478 -0.1305  0.4076 -0.6385 -0.7326  0.5855
## s.e.  0.1262  0.1271  0.1253  0.1159  0.1041  0.1611  0.3171
##
## sigma^2 = 5122169: log likelihood = -470.51
## AIC=973.01  AICc=988.56  BIC=1004.23

```

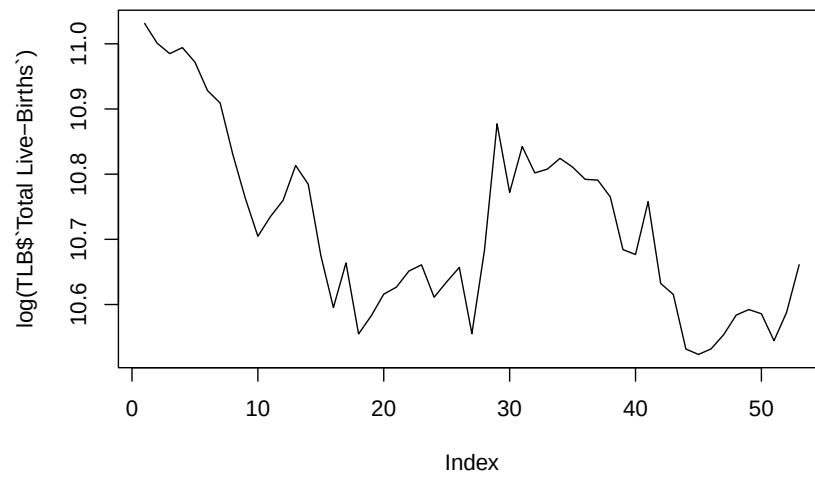
**ACF of Residuals for ARIMA(13,1,2) for TLB**



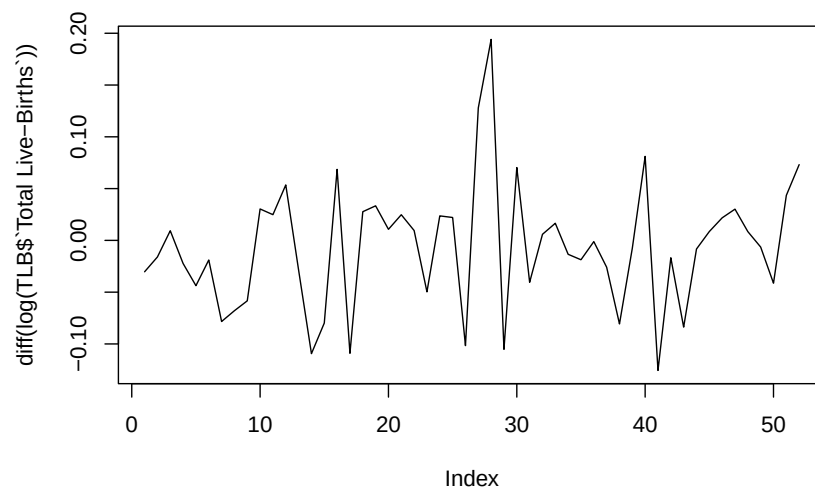
**PACF of Residuals for ARIMA(13,1,2) for TLB**



**Plot of Log Transformation for Total Live-Births (TLB) until 2012**

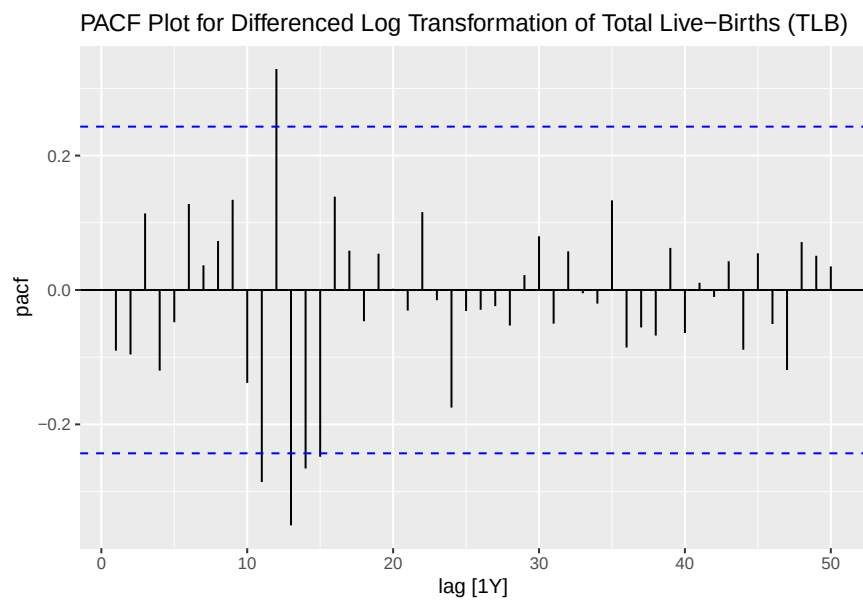
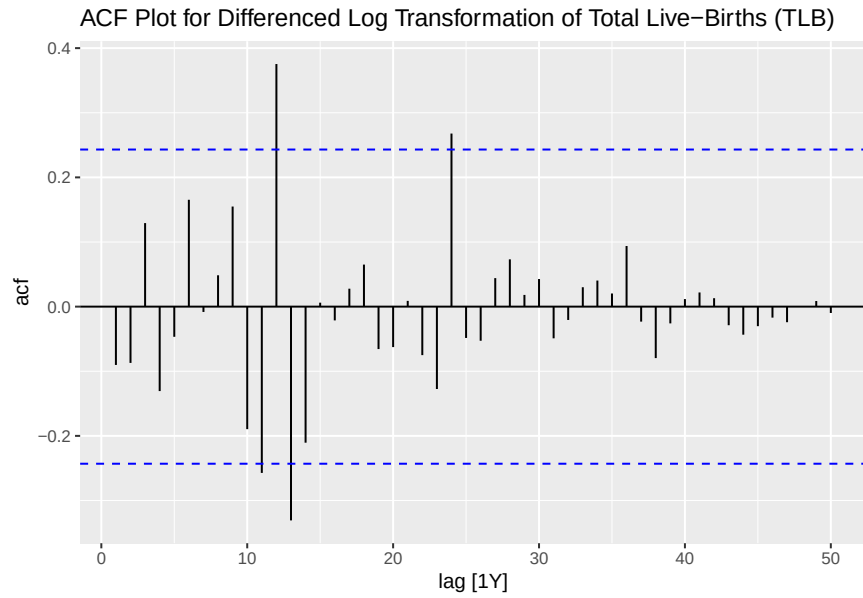


**Plot of Differenced Log Transformation for Total Live-Births (TLB) until**



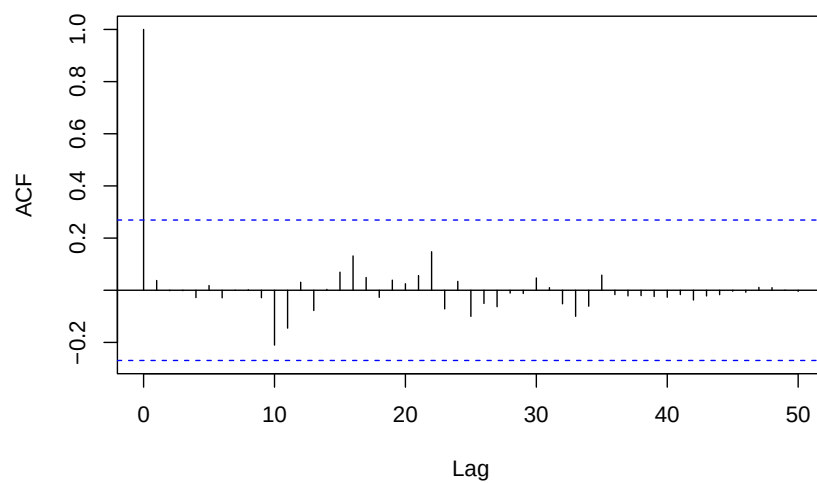
```
##      kpss_stat kpss_pvalue
##      0.1765042  0.1000000
```



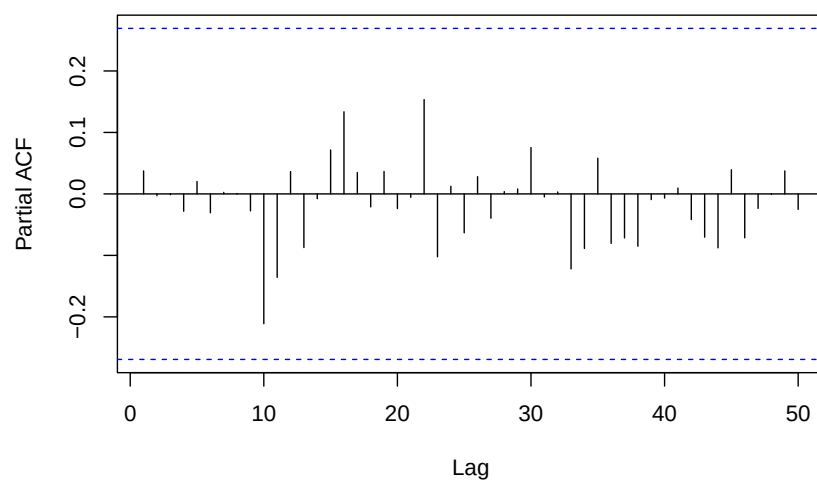


```
## Series: log(TLB_ts)
## ARIMA(15,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##    -0.1455 -0.1154  0.0919 -0.1862  0.0220  0.1559  0.1565  0.1742  0.0826
## s.e.   0.1315   0.1197  0.1028  0.0946  0.0935  0.0868  0.0972  0.0918  0.0931
##      ar10     ar11     ar12     ar13     ar14     ar15
##    -0.0706 -0.2325  0.2774 -0.3642 -0.3436 -0.4237
## s.e.   0.0939   0.0892  0.0988  0.1089  0.1264  0.1359
##
## sigma^2 = 0.00225: log likelihood = 88.78
## AIC=-145.56  AICc=-130.02  BIC=-114.34
```

**ACF of Residuals for ARIMA(15,1,0) for TLB with log transformation**



**PACF of Residuals for ARIMA(15,1,0) for TLB with log transformation**

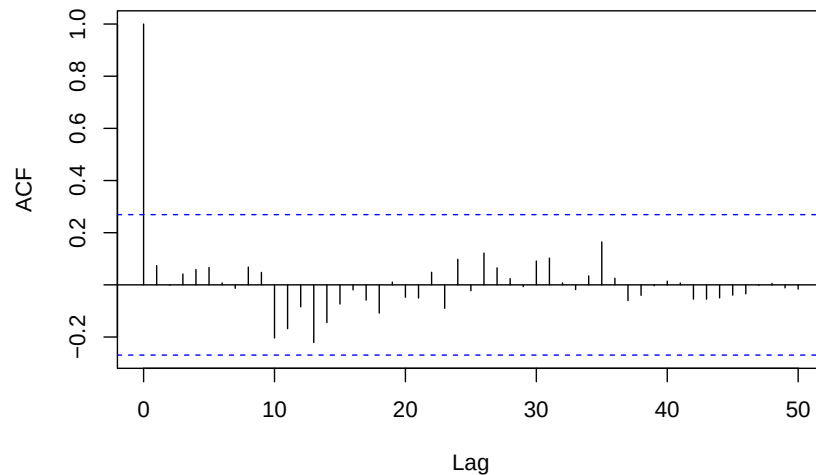


```

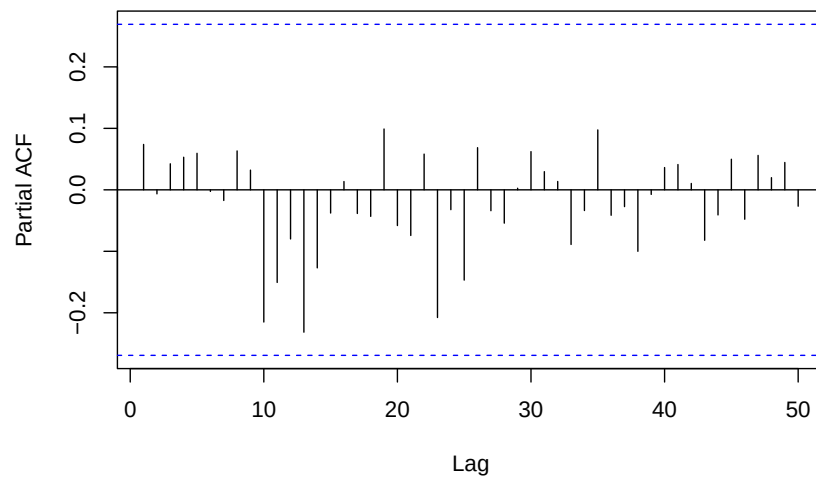
## Series: log(TLB_ts)
## ARIMA(12,1,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.3816 -0.3915 -0.4801 -0.1650 -0.0459  0.0622  0.0321  0.1705
## s.e.   0.1314   0.1535   0.1615   0.1495   0.1471   0.1454   0.1494   0.1589
##      ar9      ar10     ar11     ar12     ma1      ma2      ma3
##      0.1587  0.0805  -0.0368  0.4897  0.5443  0.4755  0.9169
## s.e.   0.1683  0.1646   0.1479  0.1451  0.1011  0.2255  0.2718
##
## sigma^2 = 0.002424: log likelihood = 85.85
## AIC=-139.69  AICc=-124.15  BIC=-108.47

```

**ACF of Residuals for ARIMA(12,1,3) for TLB with log transformation**



**PACF of Residuals for ARIMA(12,1,3) for TLB with log transformation**



## Results

| TFR Model     | AIC    | AICc   | MSE        | MAE        |
|---------------|--------|--------|------------|------------|
| ARIMA(16,1,0) | -63.99 | -45.99 | 0.04753075 | 0.1690442  |
| ARIMA(16,1,1) | -62.39 | -42.66 | 0.04783471 | 0.1700046  |
| ARIMA(15,1,1) | -60.26 | -42.26 | 0.08431368 | 0.223524   |
| ARIMA(14,1,3) | -60.99 | -40.26 | 0.01899351 | 0.09964934 |
| ARIMA(12,2,3) | -62.57 | -46.57 | 0.01218546 | 0.08274547 |
| ARIMA(11,2,3) | -52.91 | -39.19 | 0.05260825 | 0.1823268  |
| ARIMA(10,2,3) | -54.69 | -43.02 | 0.05680021 | 0.1898403  |

| Log Transformed TFR Model | AIC     | AICc    | MSE         | MAE        |
|---------------------------|---------|---------|-------------|------------|
| ARIMA(13,1,3)             | -136    | -118    | 0.02602064  | 0.121592   |
| ARIMA(12,1,4)             | -131.83 | -113.83 | 0.02101819  | 0.1132223  |
| ARIMA(13,1,5)             | -133.96 | -110.21 | 0.05826379  | 0.1840232  |
| ARIMA(12,1,5)             | -130.02 | -109.3  | 0.02172126  | 0.1147922  |
| ARIMA(15,2,0)             | -133.67 | -117.67 | 0.01869052  | 0.1056717  |
| ARIMA(15,2,2)             | -135.54 | -114.17 | 0.02235162  | 0.1187815  |
| ARIMA(14,2,2)             | -130.49 | -111.94 | 0.01087075  | 0.08352761 |
| ARIMA(13,2,2)             | -132.39 | -116.39 | 0.007426324 | 0.06826933 |
| ARIMA(12,2,2)             | -127.34 | -113.62 | 0.0498791   | 0.1720094  |

| TLB Model     | AIC    | AICc   | MSE      | MAE      |
|---------------|--------|--------|----------|----------|
| ARIMA(14,1,0) | 975.33 | 988.66 | 67367429 | 6853.008 |
| ARIMA(14,1,1) | 975.62 | 991.17 | 84179161 | 7769.997 |
| ARIMA(13,1,1) | 974.57 | 987.9  | 82856991 | 7698.072 |
| ARIMA(13,1,2) | 973.01 | 988.56 | 54382525 | 6014.802 |

| Log Transformed TLB Model | AIC     | AICc    | MSE        | MAE      |
|---------------------------|---------|---------|------------|----------|
| ARIMA(15,1,0)             | -145.56 | -130.02 | 2239721240 | 47245.34 |
| ARIMA(15,1,1)             | -145.43 | -127.43 | 2173041601 | 46542.86 |
| ARIMA(14,1,1)             | -140.56 | -125.02 | 2211919788 | 46958.53 |
| ARIMA(13,1,1)             | -141.45 | -128.11 | 2200303964 | 46836.59 |
| ARIMA(12,1,3)             | -139.69 | -124.15 | 1752591080 | 41851.92 |

## Discussion

## Conclusion

## References

Chen, M., Yip, P.S.F. & Yap, M.T. (2018). Identifying the most Influential Groups in Determining Singapore's Fertility. *Journal of Social Policy*, 47(1), 139-160. doi:10.1017/S0047279417000241.

Shair, S.N., Shaadan, N., Amirudin, N.A.B.M., & A'kashalf, N.A.M. (2024). Predictions of Malaysia Age-Specific Fertility Rates using the Lee-Carter and the Functional Data Approaches. *AIUB Journal of Science and Engineering*, 23(1), 71-78. doi: 10.53799/ajse.v23i1.1024.

Vanella, P. (2016). The Total Fertility Rate in Germany until 2040 – A Stochastic Principal Components Projection based on Age-specific Fertility Rates. *Hannover Economic Papers (HEP)*, 579. <https://hdl.handle.net/10419/155634>

## **Appendices / Statisical Appendix**

## Research Questions

Can the ARIMA model accurately forecast the Singapore's Total Live Births (TLB) and Total Fertility Rate (TFR) from 2013 to 2024 given the data from 1960 to 2012, capturing the short-term fluctuations?

## Statistical Appendix

The following statistical methods were applied throughout the analysis in the order presented below.

### Autoregressive Model (AR)

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

where:

$y_t$  is the value at time  $t$

$\phi_i$  are model parameters

$\varepsilon_t$  is white noise

Models the current values using its previous values (lags)

### Residuals

$$e_t = y_t - \hat{y}_t$$

$y_t$  is the actual observed value at time  $t$

$\hat{y}_t$  is the predicted value at time  $t$

The error which is the difference between the actual observed value and predicted value at time  $t$ , used to check the chosen model accounts for all the patterns

### Autocorrelation Function (ACF)

$$\rho_k = \frac{\text{Cov}(y_t, y_{t-k})}{\text{Var}(y_t)}$$

where:

$\rho_k$  is the autocorrelation at lag  $k$

Measures the correlation between a series  $y_t$  and lagged values  $y_{t-k}$ , which checks for trend, seasonality and whether residuals are white noise

### Box-Cox Transformation

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log(y), & \lambda = 0 \end{cases}$$

$\lambda$  was estimated using the Guerrero method

## STL Decomposition

$$y_t = T_t + R_t$$

where:

$T_t$  is the trend component,

$R_t$  is the remainder

Decomposes time series into trend and remainder components, usually with the seasonality component as well but the data in this analysis is yearly

## Moving Average (MA)

$$MA_k = \frac{1}{k} \sum_{i=-(k-1)/2}^{(k-1)/2} y_{t+i}$$

k is the window size

In this analysis when  $k = 5$ , the moving average at time t is calculated using the 2 observations before and 2 observations after t. This process smooths out short-term fluctuations in the series

<https://github.com/twalepear/TSA.git>